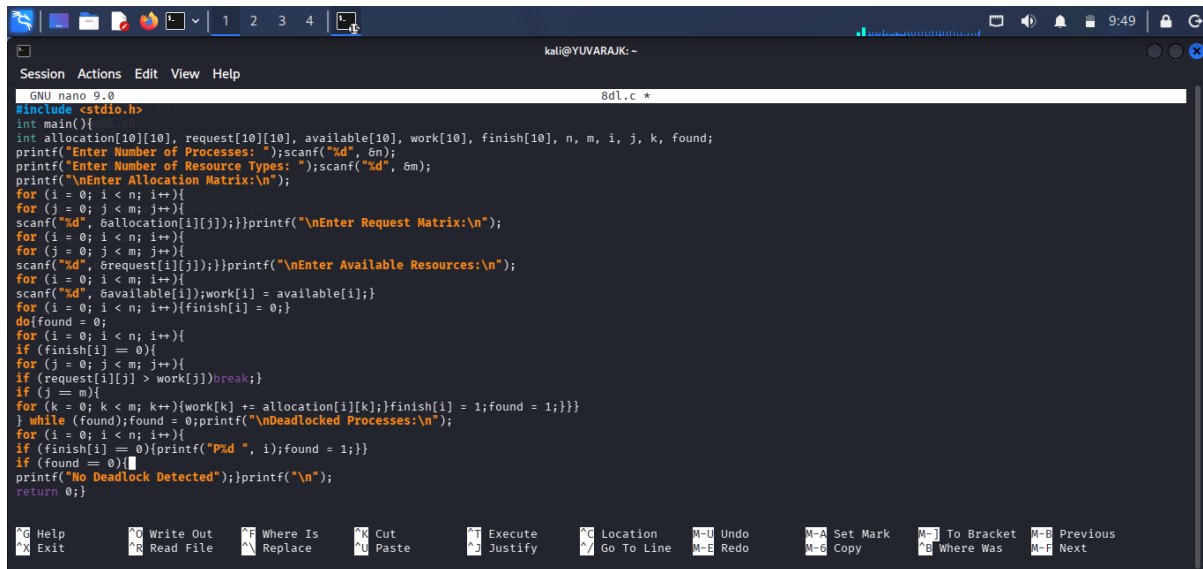
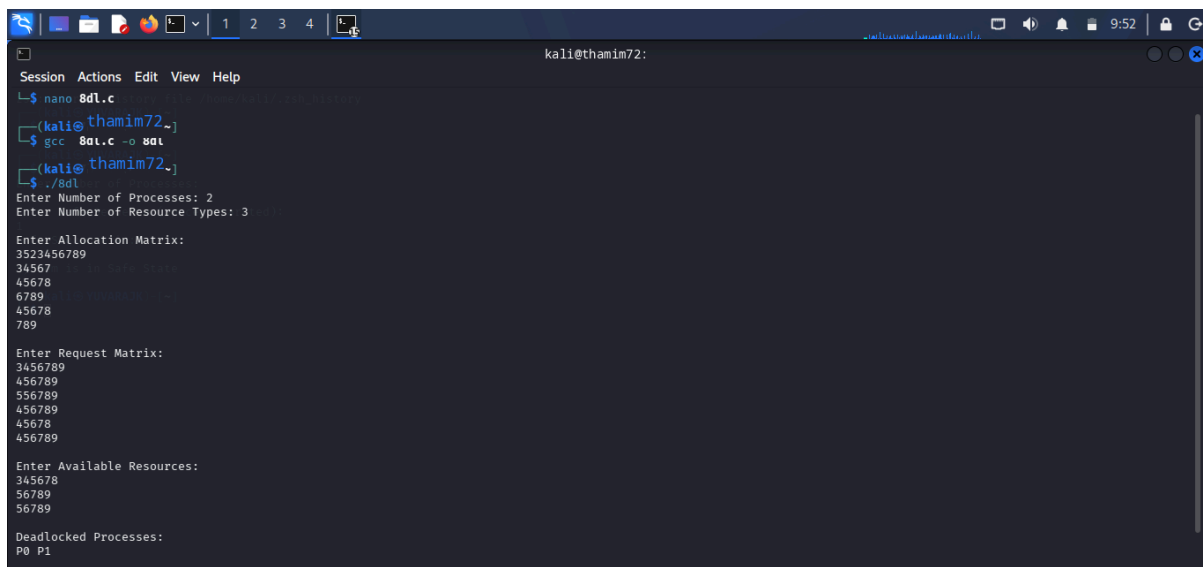


C PROGRAMMING :



```
GNU nano 9.0 8dl.c *
#include <stdio.h>
int main()
{
    int allocation[10][10], request[10][10], available[10], work[10], finish[10], n, m, i, j, k, found;
    printf("Enter Number of Processes: ");scanf("%d", &n);
    printf("Enter Number of Resource Types: ");scanf("%d", &m);
    printf("\nEnter Allocation Matrix:\n");
    for (i = 0; i < n; i++){
        for (j = 0; j < m; j++){
            scanf("%d", &allocation[i][j]);}printf("\nEnter Request Matrix:\n");
        for (i = 0; i < n; i++){
            for (j = 0; j < m; j++){
                scanf("%d", &request[i][j]);}printf("\nEnter Available Resources:\n");
            for (i = 0; i < m; i++){
                scanf("%d", &available[i]);work[i] = available[i];
                for (i = 0; i < n; i++){finish[i] = 0;
                do{found = 0;
                for (i = 0; i < n; i++){
                    if (finish[i] == 0){
                        for (j = 0; j < m; j++){
                            if (request[i][j] > work[j])break;
                            if (j == m){
                                for (k = 0; k < m; k++){work[k] += allocation[i][k];}finish[i] = 1;found = 1;}}}
                            while (found);found = 0;printf("\nDeadlocked Processes:\n");
                                for (i = 0; i < n; i++){
                                    if (finish[i] == 0){printf("P%d ", i);found = 1;}}
                                    if (found == 0){
                                        printf("No Deadlock Detected");}printf("\n");
                                        return 0;}
```

OUTPUT :



```
kali@thamim72:~$ nano 8dl.c
(kali@thamim72:~$ gcc 8dl.c -o 8dl
(kali@thamim72:~$ ./8dl
Enter Number of Processes: 2
Enter Number of Resource Types: 3

Enter Allocation Matrix:
3523456789
34567
45678
6789
45678
789

Enter Request Matrix:
3456789
456789
56789
456789
45678
456789

Enter Available Resources:
345678
56789
56789

Deadlocked Processes:
P0 P1
```

SHELL PROGRAMMING:

The screenshot shows a Kali Linux terminal window with a nano editor open. The terminal title is "kali@YUVARAJK: ~". The nano editor is editing a file named "8dl.sh". The content of the file is a bash script that prompts the user to enter the number of processes and then deadlocked process numbers. The script checks if the deadlocked process numbers are empty and either prints "No Deadlock Detected" or "Deadlocked Processes:" followed by the entered numbers. The bottom status bar of the nano editor shows "Read 12 lines" and various keyboard shortcuts.

```

GNU nano 9.0 8dl.sh
#!/bin/bash
echo "Enter number of processes:"
read n
echo "Enter deadlocked process numbers (if any):"
read processes
if [ -z "$processes" ]
then
echo "No Deadlock Detected"
else
echo "Deadlocked Processes:"
echo "$processes"
fi

```

Enter Requested Metrics:
 1:process
 2:dead
 3:dead
 4:dead
 5:dead
 6:dead
 7:dead
 8:dead

Enter Available Resources:
 1:100
 2:100
 3:100
 4:100
 5:100
 6:100
 7:100
 8:100

Deadlocked Processes:
 1:
 2:
 3:
 4:
 5:
 6:
 7:
 8:

Read 12 lines

Ctrl-H Help Ctrl-W Write Out Ctrl-S Where Is Ctrl-U Cut Ctrl-E Execute Ctrl-L Location Ctrl-M Undo Ctrl-A Set Mark Ctrl-J To Bracket Ctrl-P Previous Ctrl-X Exit Ctrl-R Read File Ctrl-W Where Is Ctrl-U Cut Ctrl-E Execute Ctrl-L Location Ctrl-M Undo Ctrl-A Set Mark Ctrl-J To Bracket Ctrl-P Previous Ctrl-B Where Was Ctrl-N Next

OUTPUT :

[illegible]